



BIT4543 ARTIFICIAL INTELLIGENCE

AI HEALTH SYMPTOM CHECKER

SYMPTOCARE

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Introduction & Problem Statement

WHY SYMPTOCARE ?

- People often search symptoms across multiple websites.
- Information can be confusing and overwhelming.
- Manually comparing symptoms is time-consuming.
- Users need a simple way to obtain an initial health assessment.
- AI can support preliminary symptom checking, but cannot replace doctors.

Objectives & Scope

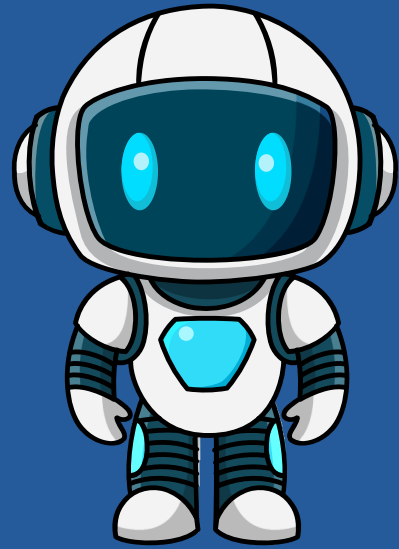
Project Objectives

1. Develop an AI-based health symptom checker.
2. Provide a simple and interactive platform.
3. Provide basic health guidance.
4. Apply AI/ML concepts to healthcare.

Project Scope

- Web-based application
- Symptom selection
- AI condition prediction
- Health guidance
- Patient demographic interface
- Input validation
- No formal medical diagnosis





Existing Solutions

Existing System	Main Limitation
Mayo Clinic	Manual decision-tree navigation
Patient.info	Lengthy/static questionnaire
WebMD	Rule-based logic
Ada Health	Multi-step intake & registration
Academic ML Models	Often limited to notebooks

Research Gap

- High onboarding friction
- Manual navigation
- Limited interactive ML interfaces
- Privacy concerns
- Lack of input validation
- Our Solution: SymptoCare
- Simple + Interactive + Privacy-First + ML-Based



Agile Development Methodology



SymptoCare was developed using an Agile approach to allow continuous testing, improvement, and integration throughout the project.

Development Process:

1. Requirements Analysis

→ Identify system requirements and healthcare domain needs.

2. Data & ML Development

→ Preprocess the dataset and train the Random Forest model.

3. Frontend Integration

→ Integrate the trained AI model into the Streamlit web application.

4. Testing & Improvement

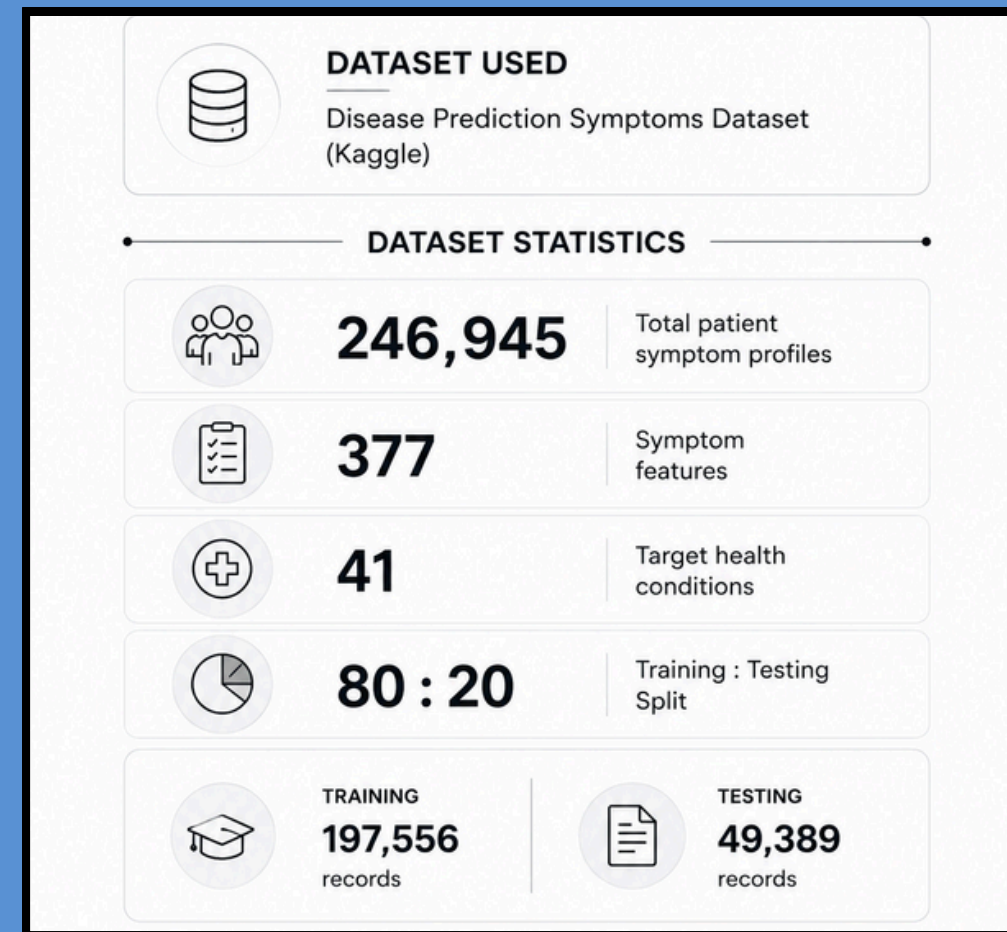
→ Test the system, identify issues, and improve the application.

Why Agile?

- Supports iterative development
- Allows continuous testing
- Makes it easier to improve the system
- Helps integrate AI and frontend components step-by-step



Dataset



Data Representation

Each symptom is represented as:

1 = Symptom Present
0 = Symptom Absent

Symptom	Input
Fever	1
Cough	0
Headache	1
Fatigue	1



Random Forest Classifier

Random Forest was selected as the main AI model because it is suitable for structured symptom data and can provide reliable prediction results.

HOW IT WORKS:

- User-selected symptoms are converted into a 377-feature binary vector.
- The vector is processed by multiple decision trees.
- Each tree contributes to the final prediction.
- The model produces the most likely health condition and its probability.

MODEL CONFIGURATION:

- 100 Decision Trees
- Gini Criterion
- 80:20 Train-Test Split
- Random State = 42



System Design & Implementation



SYSTEM WORKFLOW

The system follows a simple process to provide preliminary health assessment:

1. User Input

Users select their symptoms and may provide demographic information.

2. Input Validation

The system checks the selected symptoms before analysis.

3. Preprocessing

Selected symptoms are converted into a binary format for the AI model.

4. AI Prediction

The Random Forest model analyses the input and predicts possible health conditions.

5. Results

The system displays the primary assessment, probability breakdown, and basic health guidance.

TECHNOLOGIES USED:

Python • Streamlit • Scikit-learn • Pandas • NumPy • Pickle • Git

Testing Strategy

Unit Testing

- Model loading
- Vectorization
- Input validation

Integration Testing

- UI → Model pipeline
- Probability sorting
- Result rendering

User Acceptance Testing

- Navigation
- Usability
- Responsiveness
- Output clarity

Test Result

TC-01 → TC-07: PASS
Visual: Use a simple testing flow:
Unit → Integration → UAT → PASS



Results & Discussion

METRIC	SCORE
Accuracy	81.44%
Precision	82.0%
Recall	81.0%
F1-Score	81.0%

Key Finding

- More symptoms → Better prediction confidence
- Single symptom → lower/distributed confidence
 - ≥ 3 symptoms → stronger prediction confidence
 - Average response time $\approx$ 1.2 seconds

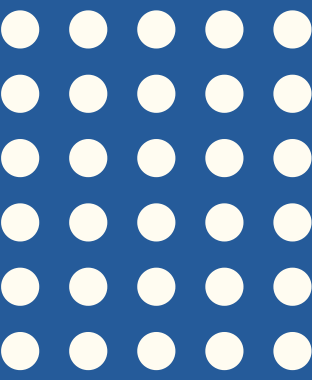
Visual:

Put 81.44% HUGE in the middle, then the other metrics around it.



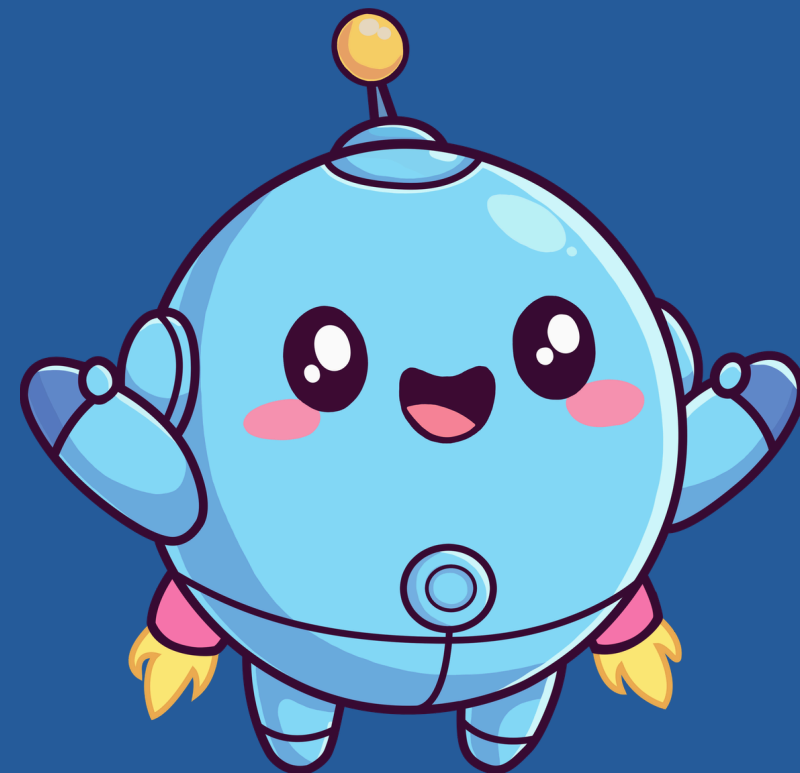


Conclusion & Future Improvements

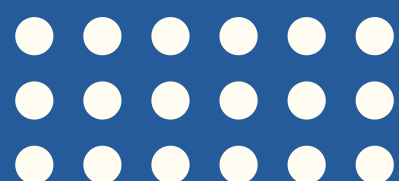
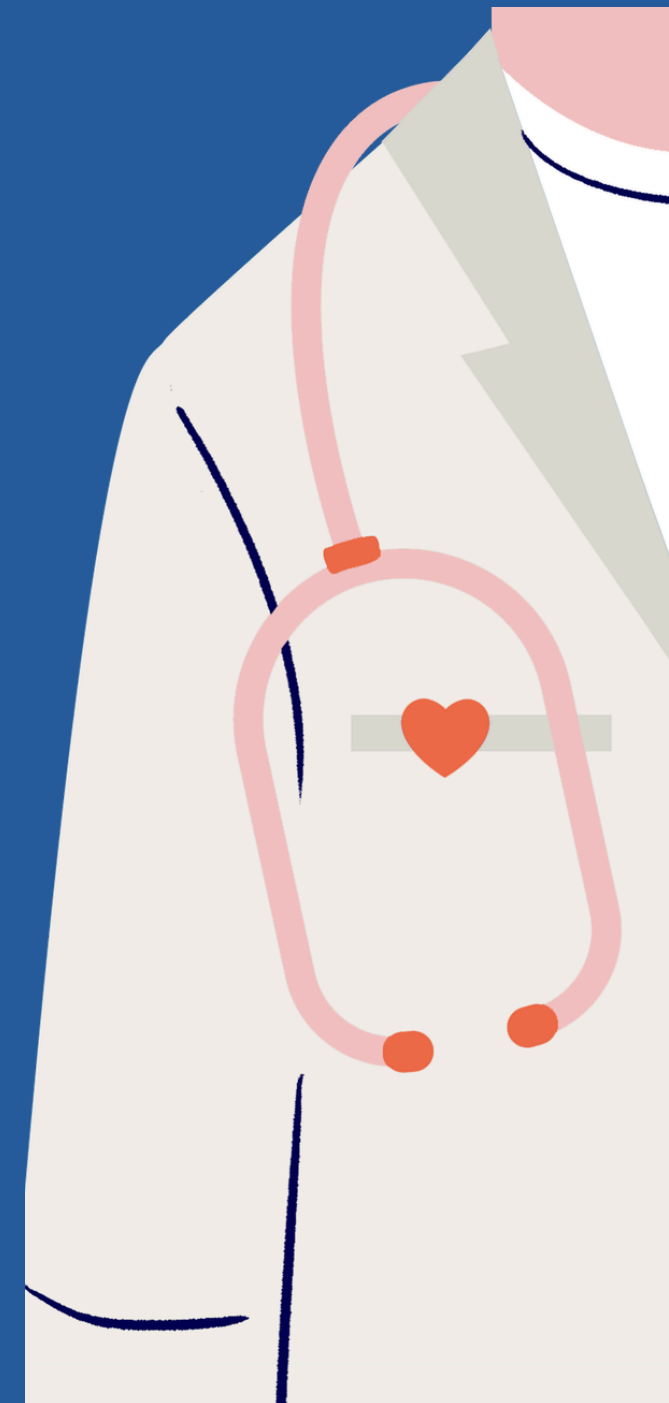


AI symptom checker successfully developed

- Random Forest integrated with Streamlit
- 81.44% test accuracy achieved
- Interactive and privacy-first interface
- Provides preliminary health information



- Larger real-world clinical datasets
- Better handling of rare conditions
- Advanced models such as XGBoost
- Cloud infrastructure scaling
- Integrate demographics into prediction
- Personalized risk assessment





LIVE SYSTEM DEMONSTRATION

Step 1: Open SymptoCare

Step 2: Go to Check Symptoms

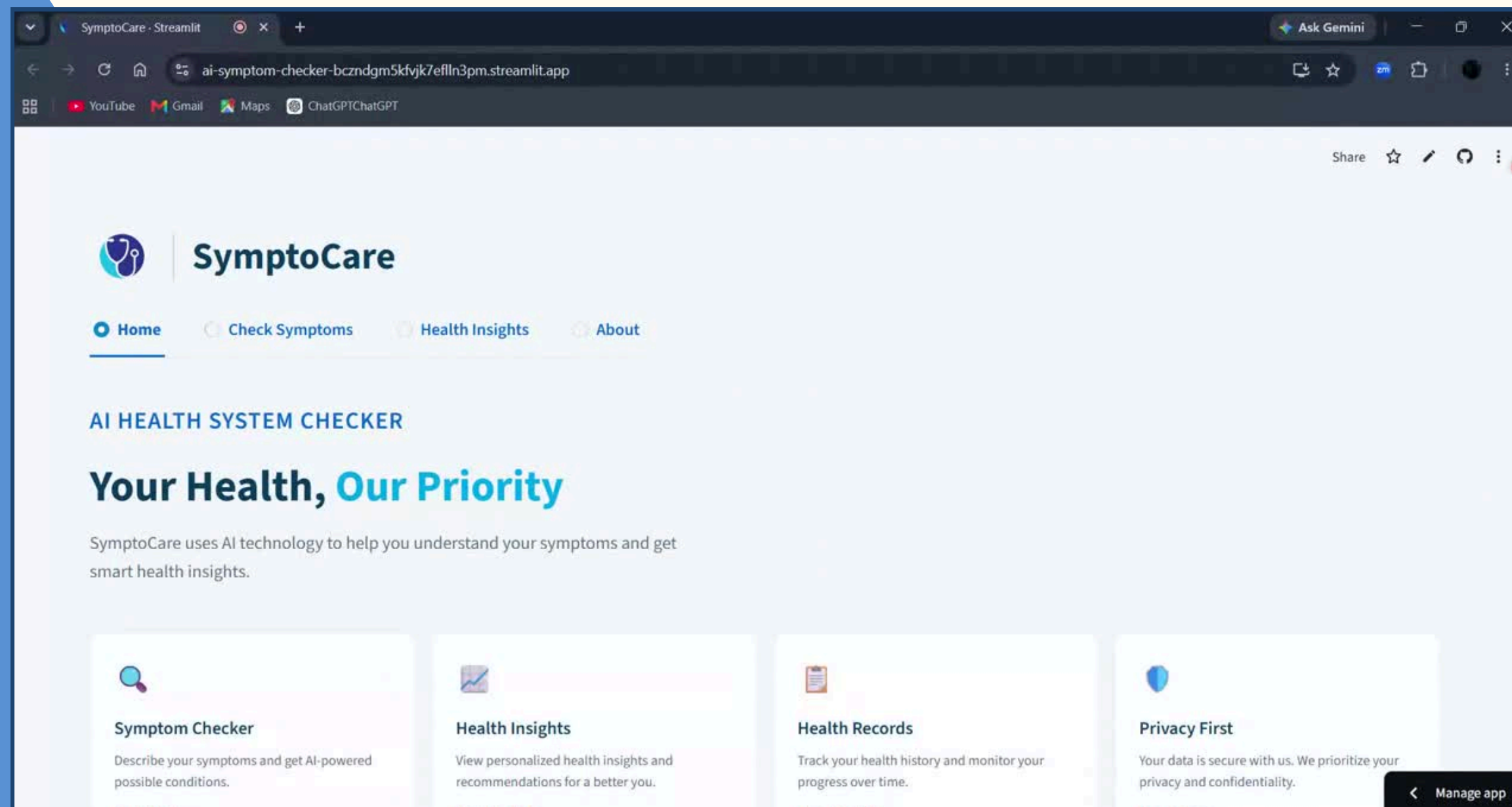
Step 3: Select symptoms

Step 4: Click Analyze

Step 5: Show:

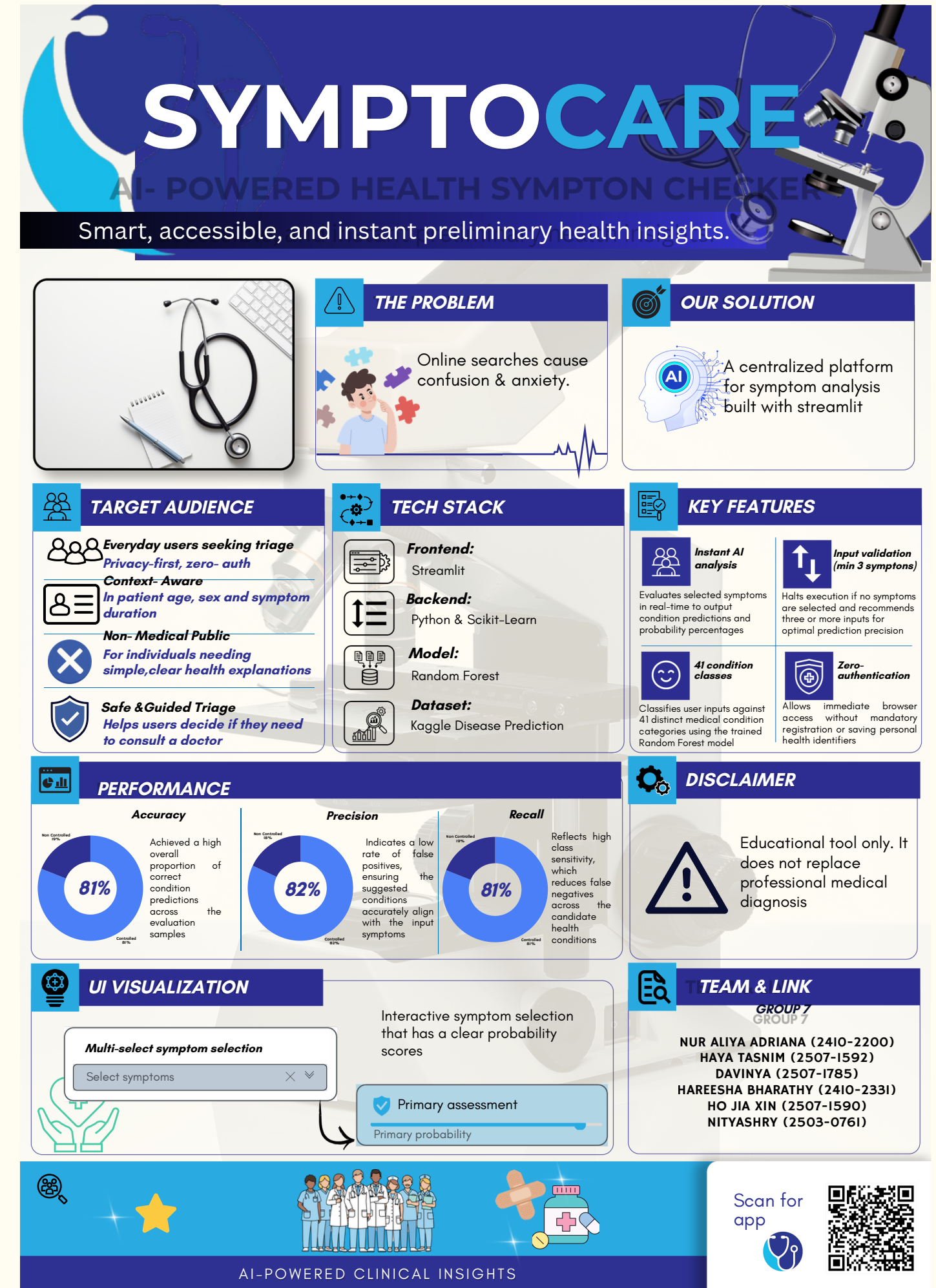
- Primary Assessment
- Possibilities Breakdown
- Percentage probabilities
- Health Guidance
- Disclaimer

Step 6: Demonstrate validation by trying to analyse without selecting symptoms.

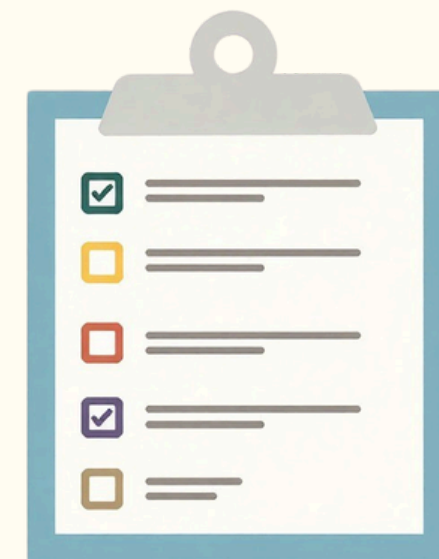




POSTER INFOGRAPHICS



Any 
Questions?





THANK YOU

**SYMPTOCARE: AI HEALTH
SYMPTOM CHECKER**